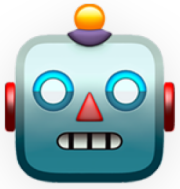
CMAA5043 — DESIGN MILESTONE CHECK-IN



BrickBuddy

LEGO Robot Building Assistant for Children

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INTERACTIVE PROTOTYPE & REPORT

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1. Project Overview

Problem Statement

Children aged 5–10 often find LEGO robot building sets overwhelming. Complex instruction booklets, small piece counts, and abstract engineering concepts can cause frustration, disengagement, and abandoned builds. There is a gap between what children *want* to create and what they feel capable of completing independently.

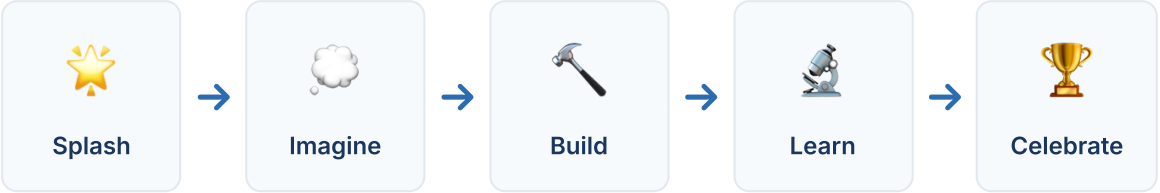
Our Solution

BrickBuddy is an AI-powered conversational assistant designed to guide children through the entire LEGO robot building experience—from imagination to celebration. It combines multimodal input (voice, camera, touch), step-by-step guided building instructions, integrated STEAM education, and emotional support into a single, child-friendly interface.

Target Users

Children aged 5–10 who are interested in building with LEGO bricks and learning about robotics and STEAM concepts. Secondary users include parents and educators who may facilitate the experience.

Interaction Flow



2. Interface Design

The prototype is organized into five distinct screens, each corresponding to a stage in the child's building journey. The visual language uses rounded shapes, warm gradients, large touch targets, and playful typography (Fredoka and Nunito font families) to create an approachable, child-friendly aesthetic.

Splash Screen

ENTRY POINT

A vibrant gradient landing page featuring the BrickBuddy mascot (animated bouncing robot), the app title, and a single call-to-action button. The design is intentionally simple to welcome young users without cognitive overload.

- Animated mascot with bounce keyframe
- Large, tactile "Let's Build!" button
- Warm orange-to-teal gradient background

Imagine Screen

STAGE 1 — IDEATION

The conversational ideation phase where children describe what they want to build. Supports three multimodal input modes, plus pre-made templates for children who prefer to browse options.

- **Talk to Me:** Voice input with animated waveform
- **Show Me:** Camera capture with viewfinder UI
- **Pick One:** Template gallery (Robot Dog, Car, Dino, etc.)
- AI understanding confirmation with chat bubbles

Build Screen

STAGE 2 — CONSTRUCTION

The core building experience with a two-panel layout: step-by-step visual instructions on the left, and a real-time chat sidebar on the right for asking questions. Each step includes a 3D-view area for the LEGO assembly, a written description with highlighted key pieces, a parts inventory display, and navigation controls to move between steps.

- Emotion support banner provides encouragement
- Numbered step progression (1 of N) with visual preview
- Color-coded piece inventory per step
- Persistent chat sidebar with text and voice input for real-time Q&A
- STEAM badges on contextual learning moments

🔬 Learn Screen

STAGE 3 — STEAM EDUCATION

An interactive Q&A section that connects the completed build to real-world STEAM concepts. Topics are filterable by discipline (Science, Technology, Engineering, Art, Math) and each answer includes fun facts.

- Discipline filter bar with color-coded tabs
- Expandable Q&A cards with fun-fact callouts
- Knowledge summary with completion checks

🏆 Celebrate Screen

COMPLETION

A rewarding conclusion featuring a trophy animation, confetti effects, build statistics (time, steps, questions asked), and a friendly prompt to share or start a new project.

- Animated confetti particle system
- Session statistics cards (time, steps, Q&A count)
- Gradient celebration background
- "Build Again" restart loop

3. Key Design Features

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Multimodal Input

Children can express their ideas through voice recording, camera capture, or template selection. This accommodates diverse age groups, literacy levels, and preference styles. The interface adapts dynamically—selecting a mode reveals its dedicated panel while keeping other options accessible.

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3-Stage Interaction Model (Imagine → Build → Learn)

The experience is structured into three pedagogical stages: creative ideation, hands-on construction, and reflective learning. A persistent progress bar across the top navigation gives children a sense of advancement, and each stage transitions smoothly with slide animations.

🔬

Integrated STEAM Education

STEAM concepts are embedded contextually throughout the build (via in-step badges and chat responses) and explored in depth on the dedicated Learn screen. Topics are organized by discipline (Science, Technology, Engineering, Art, Math) with color-coded visual cues, making abstract concepts tangible through the lens of the child's own creation.

💛

Emotional Support & Encouragement

The Build screen features an emotion support banner that provides positive reinforcement and patience-building messages. BrickBuddy's conversational tone throughout the app is warm, enthusiastic, and non-judgmental—designed to sustain motivation and reduce frustration during challenging steps.

4. Implementation

Technology Stack

The current prototype is built as a single-file interactive application using vanilla **HTML**, **CSS**, and **JavaScript**. This approach was chosen for maximum portability and ease of deployment. The file is self-contained with no external dependencies beyond Google Fonts.

Key Technical Details

ASPECT	DETAIL
Rendering	CSS-driven screen transitions with transform and opacity animations
Typography	Fredoka (headings/UI) + Nunito (body text) via Google Fonts
Layout	CSS Grid for build screen (instruction + chat panels); Flexbox elsewhere
Animations	CSS keyframes for bounce, wave, pulse, confetti-fall, and typing indicators
Interactivity	JavaScript-driven screen navigation, simulated chat, voice/camera UI, step progression
Responsiveness	Media query at 800px collapses build layout to single column

Deployment

The prototype is deployed on **GitHub Pages** via a GitHub Actions workflow for continuous deployment from the main branch.

> **Live Demo:**
> https://yusong-enceladus.github.io/CMAA5043/week9/prototype.html

5. Progress & Next Steps

What Has Been Completed

ITEM	DESCRIPTION	STATUS
Splash screen	Animated entry with mascot and CTA button	Done
Imagine screen	3 input modes (voice, camera, template), AI confirmation flow	Done
Build screen	Step-by-step instructions, piece inventory, chat sidebar, emotion banner	Done
Learn screen	STEAM Q&A with discipline filtering, fun facts, knowledge summary	Done
Celebrate screen	Confetti animation, session stats, restart loop	Done
Navigation & progress	Screen transitions, progress bar, back navigation	Done
Simulated interactions	Chat responses, voice recording UI, template selection	Done
GitHub Pages deployment	CI/CD via GitHub Actions for automatic deployment	Done

What Comes Next

ITEM	DESCRIPTION	STATUS
User testing	Conduct usability tests with 3–5 children in the target age range	Planned
Real voice input	Integrate Web Speech API for actual speech recognition	Planned
AI-generated instructions	Connect to LLM backend for dynamic, personalized build steps	Planned
3D LEGO previews	Replace emoji placeholders with Three.js or LDraw-based 3D brick renders	Planned
Camera integration	Use device camera with image recognition to identify LEGO pieces	Planned
Accessibility audit	Ensure screen reader support, keyboard navigation, and WCAG compliance	Planned
Parent/educator dashboard	Progress tracking and session history for guardians	Planned